

It is RECOMMENDED that Vendor-provided Software Update metadata, such as release note URLs, be maintained in the long-term with stable locations, preferably in a manner allowing historical caching by common online search engines, where applicable. See [Section 11.23.7.11, “ReleaseNotesUrl”](#) and [Section 11.21.2.4.8, “ReleaseNotesUrl field”](#) for sources of such information.

11.20.3.5. Transfer of OTA Software Update images

Execution of an OTA Software Update image’s transfer depends on the protocol provided in the [ImageURI](#) field of the query response.

The following are OTA Software Image transfer examples:

- An OTA Requestor invokes a [QueryImage](#) command stating only support for BDX in its [ProtocolsSupported](#). The OTA Provider, using its OTA Image Selection Logic, determines that a Software Image is available. There are several cases to be considered:
 1. The Software Image is at a URI referring to a resource on the public Internet (e.g. <https://domain.example/images/software.bin>).
 - a. The OTA Provider MAY completely download the Software Image, temporarily, to local storage. It would then reply to the OTA Requestor with a [locally-accessible BDX URI](#), such as bdx://8899AABBCCDDEEFF/software_8ce40aa1.bin. In that case, the OTA Requestor SHALL proceed with the download from the OTA Provider using the [BDX protocol](#).
 - b. The OTA Provider MAY employ a variety of buffering and proxying schemes of underlying HTTPS transfers to support the OTA Requestor downloading in real-time as a form of direct proxy. It would immediately reply to the OTA Requestor with a [locally-accessible BDX URI](#), such as bdx://8899AABBCCDDEEFF/software_8ce40aa1.bin. In that case, the OTA Requestor SHALL proceed with the download from the OTA Provider using the [BDX protocol](#). The only difference with the previous case is the fact that the transfer uses data directly proxied in real-time, as opposed to the OTA Requestor downloading a pre-stored cached copy of the same Software Image.
 2. The Software Image is already cached on the OTA Provider, either from pre-seeding over some implementation-specific scheme, or from having previously served this software update to another OTA Requestor. In that case, the OTA Provider SHALL reply to the OTA Requestor with a [locally-accessible BDX URI](#), such as bdx://8899AABBCCDDEEFF/software_8ce40aa1.bin. In that case, the OTA Requestor SHALL proceed with the download from the OTA Provider using the [BDX protocol](#).
- An OTA Requestor invokes a [QueryImage](#) command stating support for BDX and HTTPS in its [ProtocolsSupported](#). The OTA Provider, using its OTA Image Selection Logic, determines that a Software Image is available. The Software Image is at a URI referring to a resource on the public Internet (e.g. <https://domain.example/images/software.bin>). In the case of support for both HTTPS and BDX, all of the above cases are applicable, in addition to the following:
 - The OTA Provider knows that the OTA Requestor supports HTTPS from the [QueryImage](#) command. The OTA Provider MAY then respond directly to the OTA Requestor with the <https://domain.example/images/software.bin> URI. The OTA Requestor SHALL proceed to download from the public Internet using the HTTPS protocol.

As described above, an OTA Provider SHALL either proxy synchronously or asynchronously the